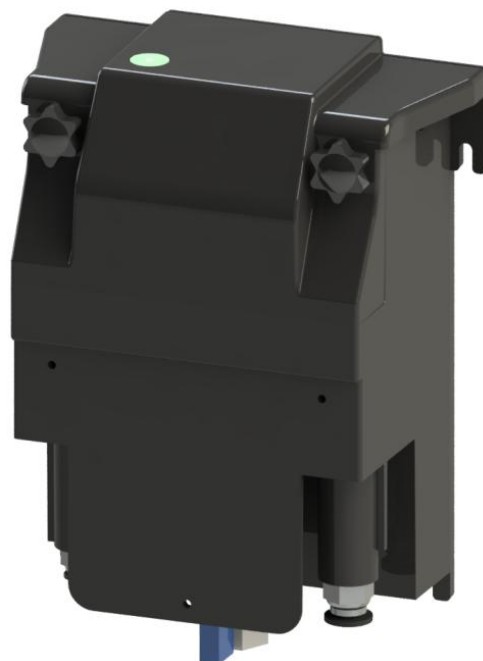


WTR01 Turbidity Sensor

Operation Manual V1.1



Caution



The sensor light source of this product may cause some damage to the eyes. Please do not point the sensor light source at your eyes.

I. Product Description

This product is a patented product with independent intellectual property rights developed by our company for online monitoring of drinking water quality. It has the characteristics of ultra-low turbidity detection limit, high-precision measurement, long-term maintenance-free equipment, water-saving operation and digital output. It can be widely used in online turbidity monitoring of tap water, secondary water supply, pipe network terminal water, direct drinking water, membrane filtered water, swimming pools, surface water, etc.

II. Basic Information

2.1 Working principle

The sensor emits a laser beam into the liquid to be measured, and the suspended particles in the liquid to be measured generate scattered light. The receiver detects the scattered light intensity in the 90° direction, and converts the scattered light intensity into a turbidity value according to the calibration table.

2.2 Product Feature

1. Sensor laser light source, higher measurement accuracy;
2. High-precision electronic devices make measurements more accurate;
3. Flow-through waterway design makes measurement more stable;
4. Unique flow channel design, waterway is not easy to accumulate dirt;
5. Integrated defoaming design to adapt to complex working conditions.

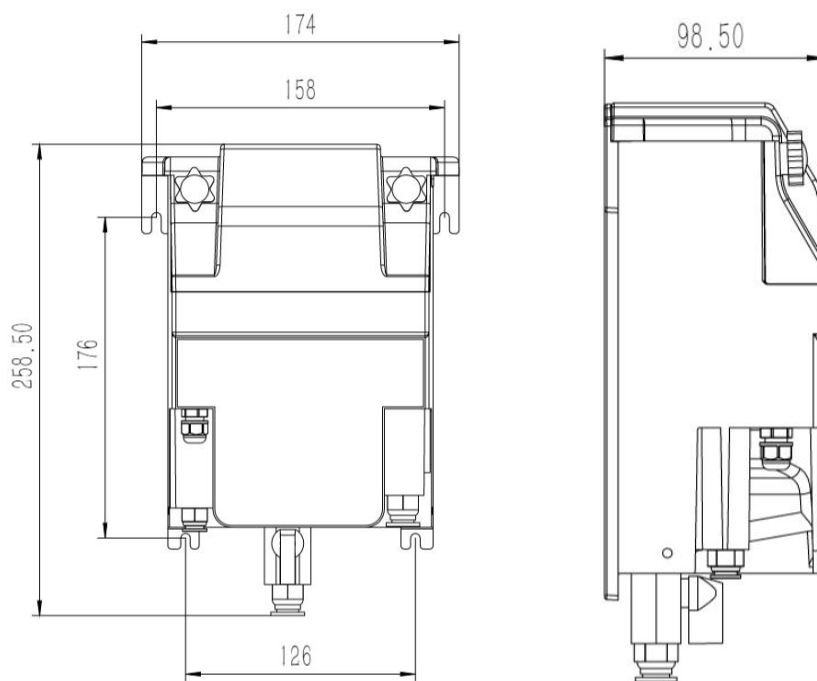
2.3 Technical Specification

The technical parameters of this product are shown in Table 2.1.

Table 2.1 Technical Parameters Table

No.	Index	Parameter
1	Power Supply	DC 24V
2	Power	5W
3	Detection Method	90° scattering
4	Light Source	660nm laser
5	Measuring Range	0-1NTU / 0-20NTU / 0-100NTU
6	Resolution	0.0001NTU(<1NTU), 0.001NTU(≥1NTU)
7	Lower Limit of Measurement	0.005 NTU
8	Accuracy	2% or 0.02 NTU (take the larger value) (based on Formazin primary standard solution at 25°C)
9	Repeatability	1% or 0.006NTU (take the larger value) (based on Formazin primary standard solution at 25°C)
10	Response Time	≤ 120 second
11	Operating Temperature	0 - 50°C
12	Protection Level	IP65 (sensor)
13	Size	174mm*259mm*99mm (width* height* thickness)
14	Material	Composite materials
15	Waterway Interface	Water inlet 6mm hose; water outlet and sewage outlet: 10mm hose
16	Inlet Water Flow	100mL /min to 900mL /min (recommend 500mL/min)
17	Communication Interface	RS485 interface

The size information of this product is shown in Picture 2.1.

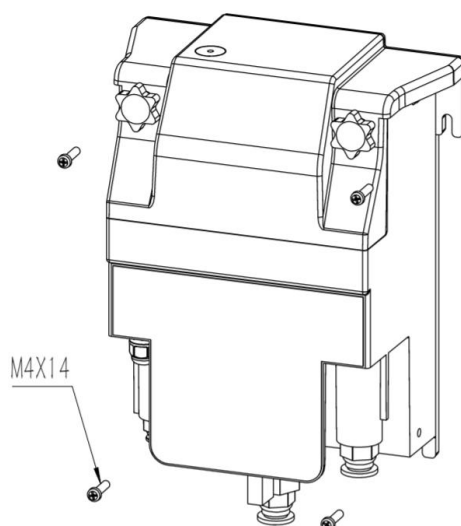


Picture 2.1 Product size

III、 Product Installation

Use four M4 screws to install the turbidity kit as shown in Picture 3.1.

Note: The equipment must be installed horizontally, as tilting will affect the measurement results.



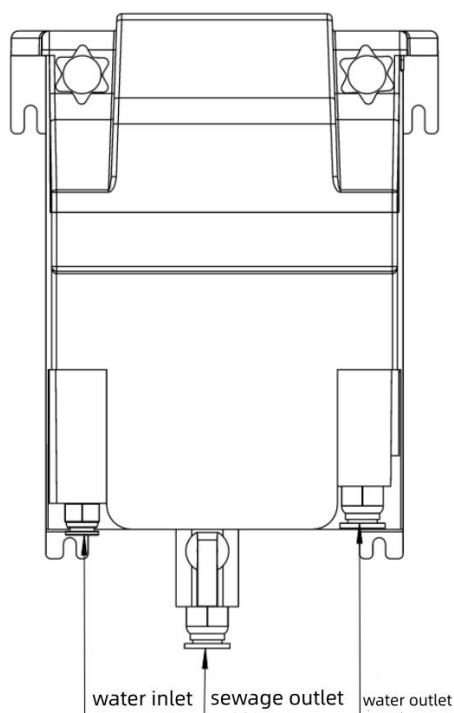
Picture 3.1 Product installation

Caution



The equipment must be installed horizontally, as tilting will affect the measurement results.

Connect the water pipe according to Picture 3.2. The recommended connection method is shown in Picture 3.3.



Picture 3.2 Water pipe location



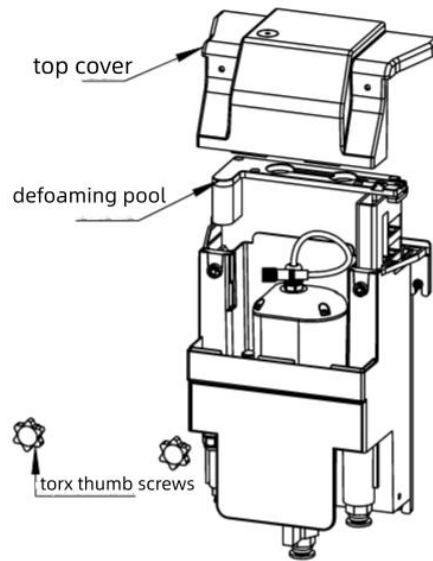
Picture 3.3 Recommended pipe connection method

Caution



If drainage is not smooth, the device may overflow even if the water inflow does not exceed the maximum flow rate!

Picture 3.4 shows the disassembly and assembly steps of the product



Picture 3.4 Product disassembly and assembly

IV. Calibration

There are two ways to calibrate the turbidity sensor: standard solution calibration and on-site calibration. It is recommended to perform maintenance before calibration, including cleaning the sensor and measuring cylinder, cleaning the sensor light source, etc. Generally, since the turbidity standard solution is not easy to obtain on site, it is recommended that users use single-point calibration during on-site calibration.

1. Standard solution calibration example

Standard solution calibration requires the preparation of low-point water standard solution and high-point water standard solution. It is generally recommended to use $<0.1\text{NTU}$ ultrapure water for low-point water. For high-point water, it is recommended to use formazine standard solution around 20NTU or range value standard solution. Turbidity values of standards can be verified using handheld devices or laboratory equipment. Calibration steps are as follows:

- 1) Cleaning sensor and measuring cylinder
- 2) Pour enough low-point water into the measuring cylinder, put the turbidity sensor into the measuring cylinder, wait for a few minutes until the average gain value of register 66 is basically stable, and then write the turbidity value and the measured average gain value of the low-point water to registers 20 and 22.
- 3) Pour out the low-point water and pour in enough high-point water, put the turbidity sensor into the measuring cylinder, wait for a few minutes until the average gain value in register 66 is basically stable, and then write the turbidity value and the

measured average gain value of the high-point water to registers 24 and 26. After both low and high points are completed, the calibration is complete.

2. On-site single point calibration

Generally, it is recommended that users use standard solutions for accurate calibration of turbidity sensors.

At the same time, in order to meet the requirements of some users who need on-site comparison and calibration, temporary calibration can also be performed on-site. The operation process is as follows: after waiting for the turbidity sensor measurement to stabilize, and the average gain value in register 66 is basically stable, write the turbidity value that needs to be calibrated (in mNTU) to register 36. After waiting for a few seconds, the turbidity sensor completes the single-point automatic calibration.

Precautions:

- 1) Since the calibration point is the slope determination point, it is recommended that the turbidity of the on-site comparison water be not less than 0.5NTU or not less than 80% of the average operating turbidity value. For example: the average turbidity value of the field water is 0.4NTU, and it cannot be lower than $0.4 \times 80\% = 0.32\text{NTU}$ during comparison and calibration;
- 2) During the comparison and calibration process, water quality turbidity should not change too much, and it is recommended that the change be kept within 2%;
- 3) The accuracy of comparison calibration depends on the change in water turbidity during the calibration process, the accuracy of the comparison calibration equipment used, and the standardization of the operator's operation of the comparison equipment.

VI. Maintenance

1. Regularly check whether the water supply and drainage are normal.
2. Regularly clean the sensor light window and the inside of the measuring cylinder.
3. Regularly compare the measurement accuracy. If the measurement error exceeds the requirements, the sensor needs to be recalibrated.

Caution	
	Regarding the problem that the measured value is too large when measuring turbidity below 0.1NTU or at low turbidity. Reason one: the flow cell has not been cleaned for a long time. Treatment method: thoroughly clean the flow cell. Reason two: low point calibration standards are inconsistent. Treatment method: use standard unified comparison equipment to recalibrate the low point.

VII. Trouble Analysis and Troubleshooting

If you encounter a fault, please try the solutions listed in Table 7.1.

Table 7.1 Failure analysis and treatment

Fault phenomenon	Possible cause analysis	Troubleshooting
RS485 cannot communicate	1. The signal cable is not connected properly 2. Sensor damaged	1. Reconnect the signal cable 2. Contact us
Abnormal increase in turbidity value (excluding water source problems)	1. Improper installation causes water contamination of the sensor optical hole 2. The bubbles are too large and exceed the defoaming capacity of the equipment 3. Contamination inside the equipment	1. Clean the sensor optical hole glass 2. Add an external defoaming device to improve defoaming capability; Or reduce the water inlet volume to less than 300ml/min 3. Clean the internal water line and sensor of the equipment (operators must undergo relevant operating training)
Turbidity value too low	Light source damaged	Contact us

Equipment leaks	1. Excessive water ingress 2. Poor drainage 3. The device is clogged inside	1. The water inflow is controlled within 900mL/min 2. The water outlet and sewage outlet hoses should be controlled within 20cm and do not use the same hose 3. Clean the inside of the device
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